$$D_1 = t_0 + a_C C + a_S S + a_T (T - T_{ref}) + a_V (V_{ref}/V)^p$$

$$D_2 = D_1 + a_C C[(V_{ref}/V)^q - 1] + a_{CT} C(T - T_{ref}) + a_{CS} CS$$

$$I_1 = I_0 \, e^{b_T (T - T_{ref}) + b_V (V - V_{ref})}$$

$$I_2 = I_0 \, e^{b_T \Delta T + b_V \Delta V + b_{TT} \Delta T^2 + b_{VT} \Delta T \Delta V}$$

$$s_i = T_{req} - T_{arr} - N D_i - D_{wire}, \quad P_i = V I_i$$

$$g_i = \min(s_i,\ k(P_{cap} - P_i))$$